PRACTICAL LAB REPORT

WEEK 1 | PROJECT MODULE 1

Setup of a Cybersecurity Testing Lab Using VirtualBox and Kali Linux

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Platform: Oracle VirtualBox 7.2.22 **Guest OS:** Kali Linux

1. Objective

The purpose of this practical was to prepare a virtual cybersecurity lab using VirtualBox and Kali Linux. The work included creating a virtual machine, configuring its network, installing the guest operating system, checking its IP configuration, and setting up access to a host Downloads folder.

2. Lab Requirements

The supplied handout specifies VirtualBox, Kali Linux as a virtual machine, a NAT Network on the 10.0.0.0/24 subnet, clipboard and drag-and-drop support, a shared host Downloads folder, Kali IPv4 address 10.0.0.2/24, Internet access, and a final virtual-machine snapshot.

3. Tools Used

- Windows host computer
- Oracle VirtualBox 7.2.22
- Kali Linux virtual machine
- VirtualBox NAT Network and Shared Folders settings

01 | VirtualBox installation

Oracle VirtualBox 7.2.22 was installed on the Windows host. After installation, VirtualBox Manager was opened to confirm that the virtualization software was ready for the lab configuration.

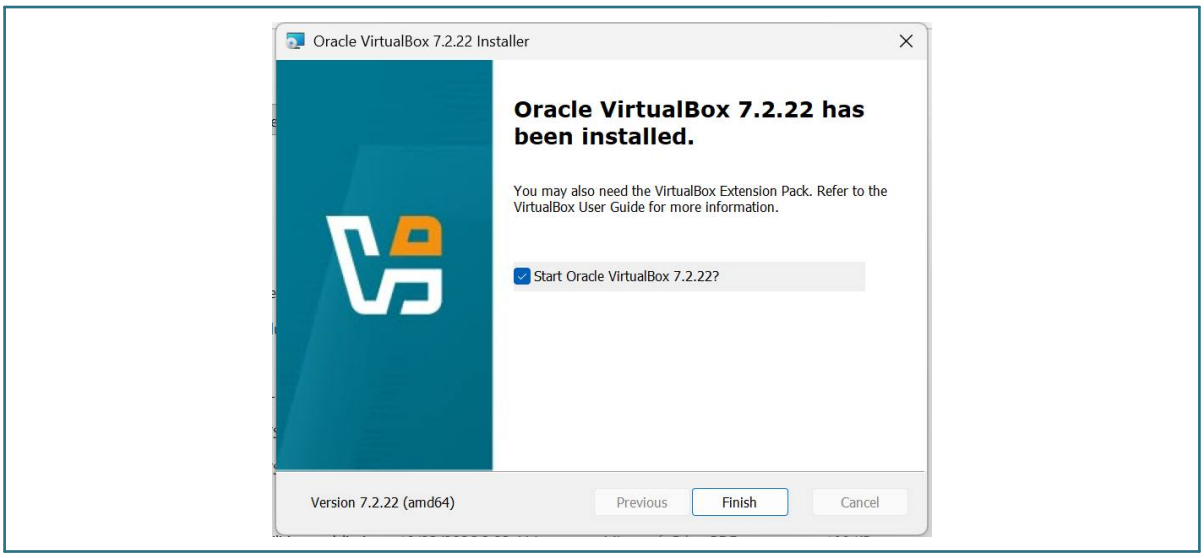


Figure: VirtualBox installation completion.

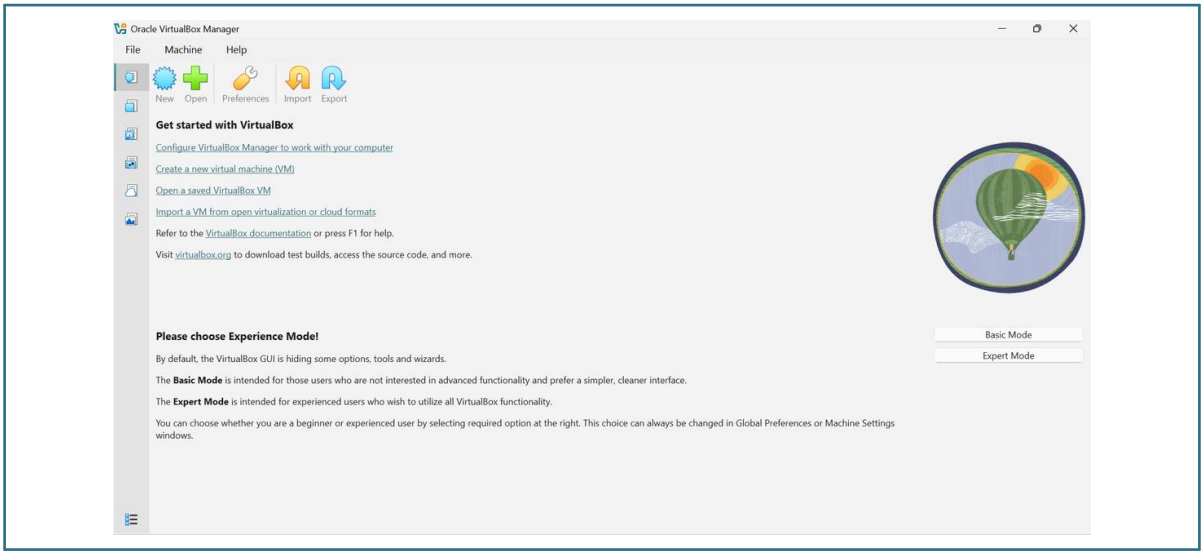


Figure: VirtualBox Manager launched.

02 | Creating the Kali virtual machine

A new guest machine named KALI was defined in VirtualBox. The setup process included selecting the Kali Linux installation media and assigning the virtual machine's basic hardware and storage resources.

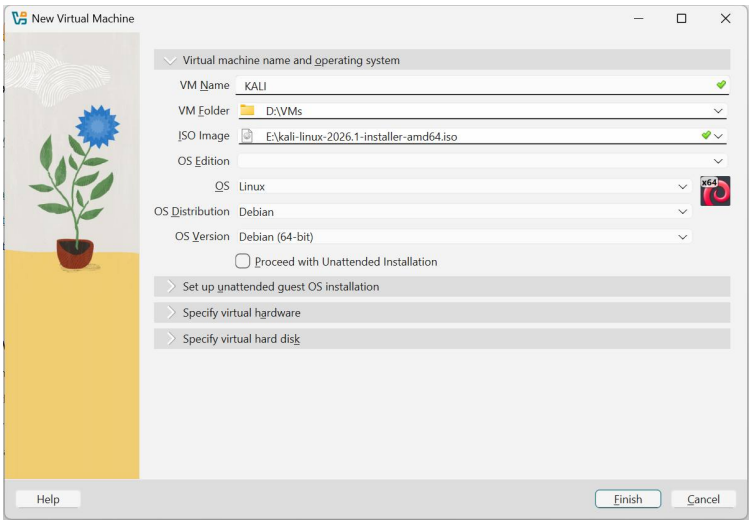


Figure: Kali VM setup wizard.

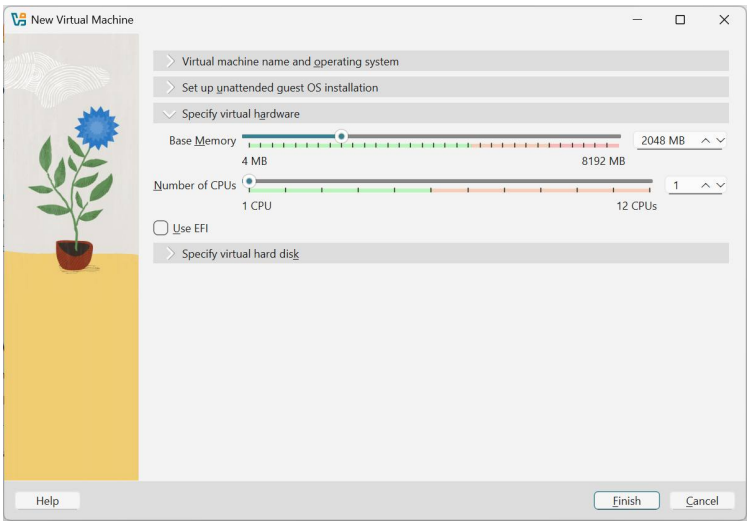


Figure: VM resource configuration.

03 | Virtual network settings

The KALI machine was connected through the NatNetwork adapter. This is the NAT Network arrangement required by the practical, with the lab using the 10.0.0.0/24 address range.

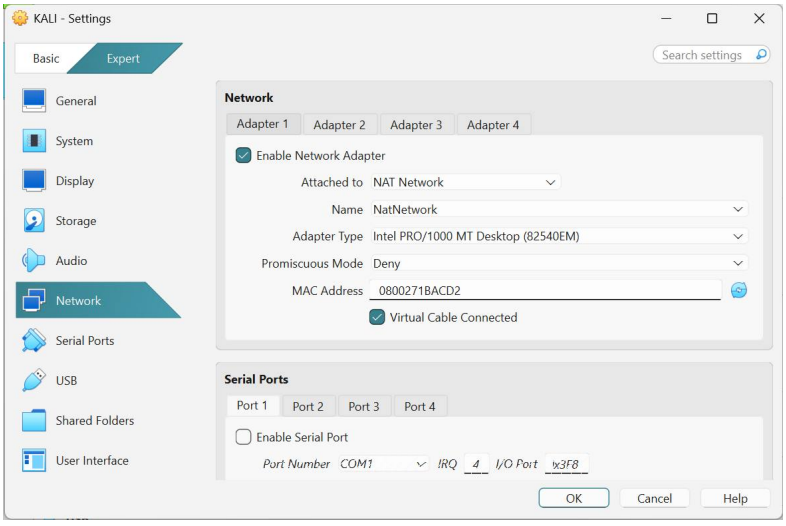


Figure: NatNetwork selected.

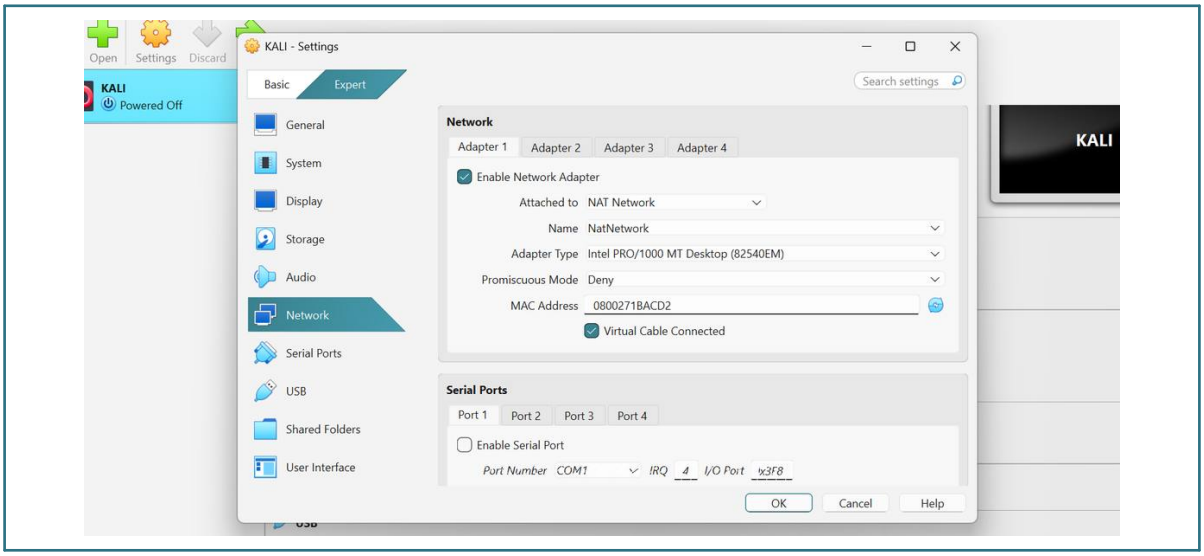


Figure: Adapter configuration.

04 | Kali Linux installation

The virtual machine was booted from the Kali Linux installation media. The screenshots record the installer starting and moving into its initial network setup stage.

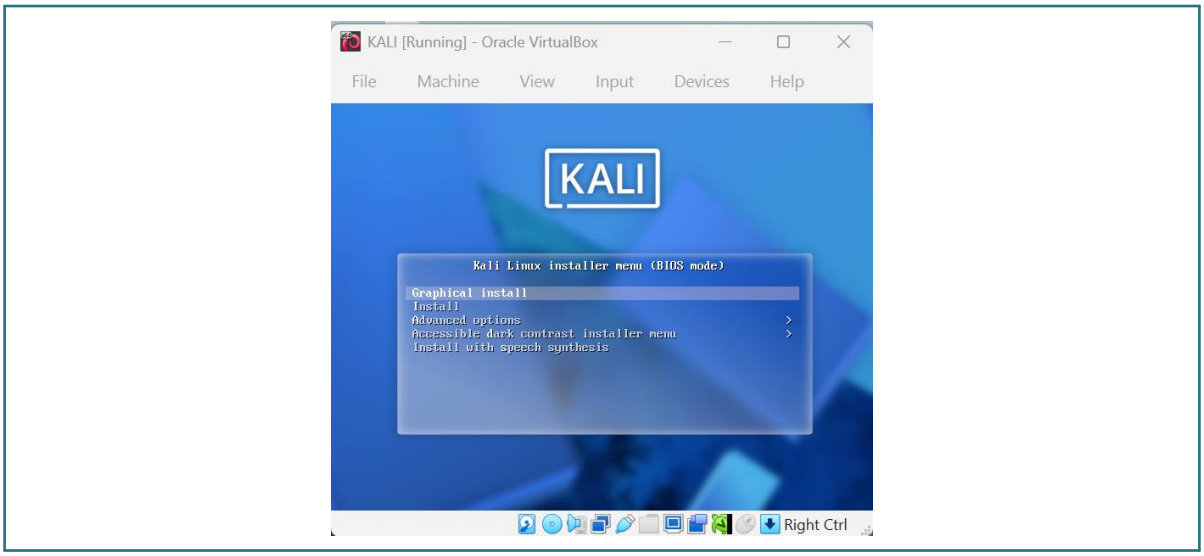


Figure: Kali installer start screen.

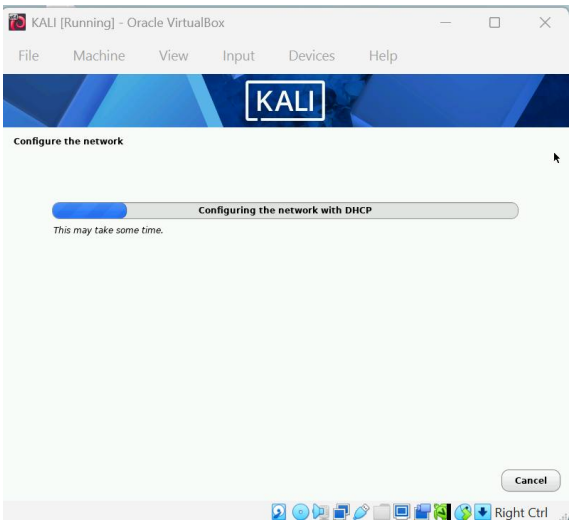


Figure: Installer network setup.

05 | IP address and connectivity checks

The Kali interface was configured with the required 10.0.0.2/24 address. A terminal ping test was then performed and the captured output shows 15 replies received with no packet loss. The network status menu also shows the wired connection active. Together, these screenshots document the guest's network configuration and connectivity check.

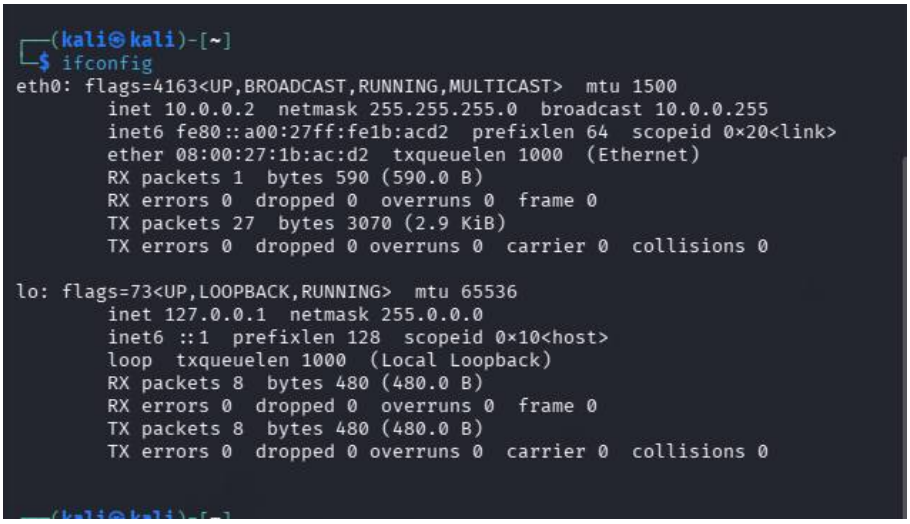


Figure: Kali interface output showing 10.0.0.2/24.

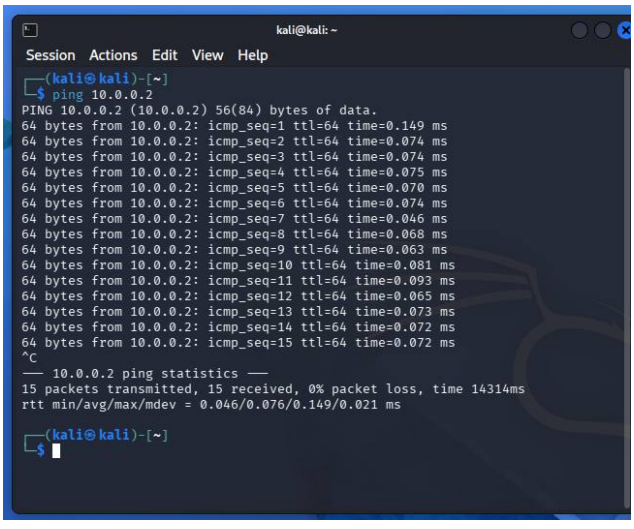


Figure: Ping test output: 15 packets received, 0% packet loss.

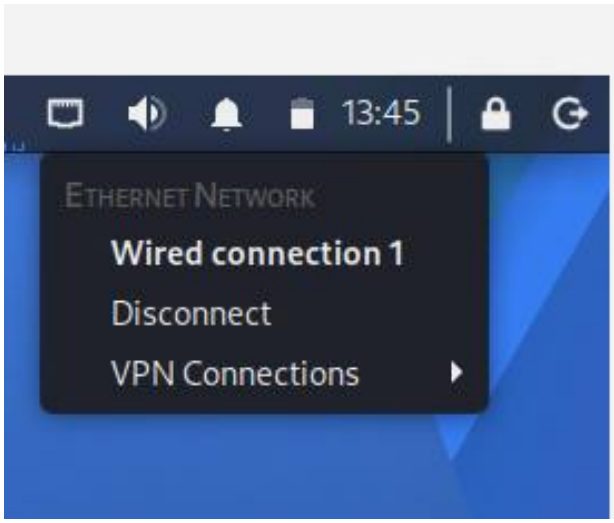


Figure: Kali wired network connection status.

06 | Host Downloads folder sharing

The host Downloads directory was selected in VirtualBox's Shared Folders configuration, with Auto-mount enabled. The screenshots show the configuration under the anonymous host profile and the Downloads share available from the Kali desktop.

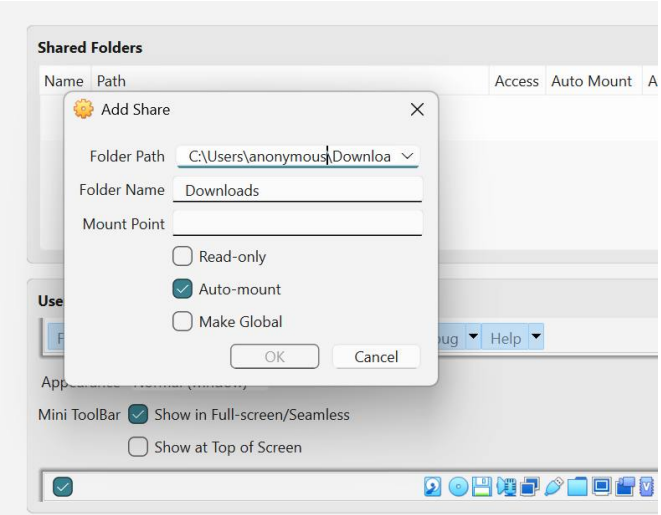


Figure: Shared folder settings under the anonymous host profile.

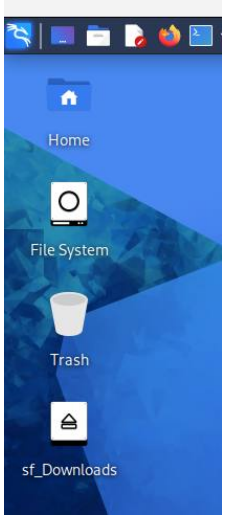


Figure: Shared Downloads entry on the Kali desktop.

07 | Final validation and snapshot

The lab configuration was completed successfully. Clipboard sharing and drag-and-drop were enabled in the virtual-machine settings, the Kali guest was confirmed to have Internet access, and a VM snapshot was created as a recovery point. These final checks completed the requirements listed in the practical task.

Conclusion

The VirtualBox-based Kali Linux lab was set up and validated in line with the practical requirements. The Kali guest was configured on the required NAT Network with address 10.0.0.2/24, the host Downloads folder was shared, and the supporting integration settings were enabled. Internet access was confirmed and a VM snapshot was saved. The assigned lab task was completed.